

Mostafa Abdo

Senior Backend Engineer — distributed systems, data platforms & AI-agent tooling

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Backend engineer with 10+ years shipping distributed systems in Python and Java — microservices, multi-tenant APIs and data pipelines for logistics, data-platform and fintech products. Currently building the MCP connector and open-source agent plugins that let Claude, ChatGPT, Codex and Cursor price, quote, book and track international freight at Freight Right; previously Git-native data workflows at Y42 in Berlin. Founder of Zolvio, an AI conversion-optimization app on the Shopify App Store. Owns systems from design through production.

10+ years shipping backend systems	16 MCP tools live in production	6 AI clients: Claude Code, Codex, Copilot, Cursor, Grok, Muse	17/17 behavioural eval cases at the release gate
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EXPERIENCE

Software Engineer · Freight Right

Jan 2025 – Present · Los Angeles, CA

FMC-licensed international freight forwarder — air, ocean, trucking and customs across 180+ countries; Shipment Manager platform, rates API and instant online booking.

- Built the Freight Right MCP connector — **16 OAuth 2.1 + PKCE-secured tools** over streamable HTTP that let Claude, ChatGPT, Codex and Cursor price, quote, prepare bookings and track FCL, LCL, air, LTL and FTL freight; live at mcp.freightright.com.
- Designed its safety model: read-only by default, bookings only ever *prepared* as a link the customer confirms in-browser, and price checks that dedupe identical requests instead of re-spending the monthly allowance.
- Published the open-source agent plugin ([freight-right/agent-plugins](https://freight-right.com/agent-plugins)): five workflow skills, native manifests for six AI clients, a contract snapshot that stops skills drifting from the real tool surface, and a **17-case behavioural eval suite** gating every release — 17/17 at 1.00.
- Architected Python/Django rate orchestration across **8 carrier providers** for air, ocean and trucking, powering instant quotes.
- Automated Shopify-to-CargoWise fulfillment and shipment tracking across 6 freight providers, and cut Shopify catalog pagination calls by **up to 80%** with larger GraphQL batches and streamed ingestion.
- Delivered Java/Spring Boot shipment APIs with tenant isolation, scoped tokens and rate limits.

Software Engineer · Y42

Feb 2022 – Dec 2024 · Berlin, Germany

Turnkey data-orchestration platform — dbt, Airbyte and Git-based workflows on Snowflake and BigQuery; \$31M Series A.

- Launched a multi-workspace Spaces API, isolating client projects and cutting onboarding time by **40%**.
- Architected a Git proxy service so teams commit, pull and push to their own Git provider from the UI, and unified the GitHub, GitLab and Bitbucket APIs behind one integration layer.
- Built a pull-request service enabling branch reviews and approvals before changes reach production.
- Scaled the platform with access control, rate limiting, dbt pipeline integration and an LLM assistant.

EXPERIENCE, CONTINUED

Software / Data Engineer · Proactive Solutions

Feb 2021 – Jan 2022
Riyadh, Saudi Arabia

- Engineered FastAPI/Starlette backend services integrating stock-trading apps, improving data sync by **15%**.
- Optimized Databricks pipelines ingesting trading data (processing time **–20%**) and built Azure Data Factory ETL that cut manual handling by **30%**.
- Shipped ML forecasting models and Dash dashboards for commodity-market data used by traders.

Full Stack Developer · Expert

Dec 2018 – Jan 2021
Riyadh, Saudi Arabia

- Led Open-Buffer, a modular SaaS (Flask, PostgreSQL, React), cutting client onboarding by **20%**.
- Architected a mobile backend (Flask, MongoDB) with GPS tracking integrated into the company ERP.
- Shipped a document-management system with OCR, archiving and search for law-firm case management and billing.

Backend Developer · IT Systems Corporation

Jun 2017 – Nov 2018
Cairo, Egypt

- Integrated Stripe and PayPal into a game-store platform handling **100K+ daily transactions**.
- Built core-banking integration between POS terminals and local banks, cutting processing time by 20%.
- Resolved critical bugs in a legacy ERP and migrated client data to the updated version.

Software Engineering Intern · Hintegration

Jun 2016 – May 2017
Cairo, Egypt

- Hardened and modernized a Python/Django CRM used by banks — secure client-data entry, faster queries, SQL-injection protection and email notifications.

EDUCATION

M.S. Computer Science, Data Analytics concentration

2025

Westcliff University · Irvine, CA

Postgraduate Studies, Machine Learning

2021

Information Technology Institute (ITI) · Cairo, Egypt

B.S. Computer Science

2016

Fayoum University · Egypt

INDEPENDENT PRODUCT

Zolvio · Founder & engineer

zolvio.io · Shopify App Store ·
launched Jan 2026

AI conversion-optimization app for Shopify merchants: automated store audits (43 built-in checks plus AI screenshot analysis), session recordings, heatmaps, surveys and A/B tests, with theme-specific fix instructions. Built on Claude models; free tier through a \$249/month Pro plan.

SKILLS

LANGUAGES & FRAMEWORKS

Python, Java, Django, Spring Boot, FastAPI, Flask, GraphQL, React, Dash, Pandas, scikit-learn

SYSTEMS

Distributed systems, microservices, REST APIs, multi-tenancy, OAuth 2.1, rate limiting, MCP and AI-agent tooling, agent evals

DATA & STORAGE

PostgreSQL, MySQL, MongoDB, Redis, Snowflake, BigQuery, Databricks, dbt, Azure Data Factory

CLOUD & DEVOPS

AWS (EC2, RDS, Lambda, S3), GCP (GKE, GCS), Docker, Kubernetes, Terraform, Kafka, GitHub Actions

INTEGRATIONS

Shopify, CargoWise, GitHub / GitLab / Bitbucket APIs, Stripe, PayPal

LINKS

github.com/freight-right/agent-plugins — Freight Right agent plugins (MIT)

apps.shopify.com/zolvio — Zolvio on the Shopify App Store

ronozoro.github.io — portfolio